

A. Hello World program.

- • create a class Main
- print Hello, Java World!

⇒ Source code

```
public class Main {  
    public static void main (String[] args) {  
        System.out.print ("Hello, Java World!");  
    }  
}
```

Output

→ Hello, Java World!

B. Variables & Data Types

Write a program demonstrating:

- int, double, boolean, char, String
- print their values.

Source code

```
Public class DataTypes {  
    Public static void main (String [] args) {  
        int num = 10;  
        double dub = 1.98245;  
        boolean bool = true;  
        char ch = 'A';  
        String str = "I need Full Marks";  
        System.out.println (num);  
        System.out.println (dub);  
        System.out.println (bool);  
        System.out.println (str);  
        System.out.println (ch);  
    }  
}
```

Output

```
10  
1.98245  
true  
I need Full Marks  
A
```

c) Operators program

write a java program showing:

- Arithmetic operators (+, -, *, /, %)

→ Source code

```
public class Arithmetic {  
    public static void main (String [] args) {  
        int a = 100;  
        int b = 5;  
        int add = a + b;  
        int subtract = a - b;  
        int multiply = a * b;  
        int divide = a / b;  
        int remainder = b % a;  
        System.out.println (add);  
        System.out.println (subtract);  
        System.out.println (multiply);  
        System.out.println (divide);  
        System.out.println (remainder);  
    }  
}
```

output

105

95

500

20

5

- Relational operators

→ Source code

```
Public class Relational {  
    Public static void main (String[] args) {  
        int a = 200;  
        int b = 150;  
int c = 100;  
        System.out.println (a + " is greater than " + b);  
        System.out.println (a > b);  
        System.out.println (a < b);  
        System.out.println (a == b);  
        System.out.println (a != b);  
    }  
}
```

Output

200 is greater than 150
true
false
false
true

• Logical operators (&&, ||, !)

→ Source code

```
public class Logical {
```

```
    public static void main (String[] args) {
```

```
        System.out.println ((100 > 130) && (460 > 500));
```

```
        System.out.println ((180 > 130) && (460 < 500));
```

```
        System.out.println ((240 < 200) || (90 > 70));
```

```
        System.out.println ((240 > 200) || (90 < 70));
```

```
        System.out.println ((240 < 200) || (90 < 70));
```

```
        System.out.println (!(120 == 80));
```

```
        System.out.println (!(120 > 800));
```

```
    }
```

```
}
```

Output

False

true

true

true

false

true

true

D User Input program

Use Scanner to take two numbers and print their Sum.

→ Source code

```
import java.util.Scanner;  
public class userInput {  
    public static void main (String[] args) {  
        Scanner in = new Scanner (System.in);  
  
        System.out.println ("Enter first number: ");  
        int num1 = in.nextInt ();  
        System.out.println ("Enter second number:");  
        int num2 = in.nextInt ();  
        int sum = num1+num2  
        System.out.println (sum);  
        in.close();  
    }  
}
```

Output

Enter first number :

10

Enter second number:

20

30